Supplementary Material: Graph-Guided Mixture-of-Experts for Unified Multimodal Mental Health Assessment

Anonymous submission

Experimental Pipeline Overview

The graph-routing investigation is one of the paper's supporting analyses (the main empirical question is the construct-bottleneck result, main paper Section 4). To isolate the routing effect, we compare five graph-routing ablation variants (V0–V4) against a non-graph baseline, across the 5 canonical seeds. We deployed a homophily diagnostic comparing real graph edges against random pairing, alongside two single-seed exploratory fixes: a K-sensitivity sweep ($K = 5, 10, 15, 20$) and learning a per-task λ weight instead of fixing it at 0.5. Under 5-seed, reportability-gated statistics, routing shows no distinguishable effect on DAIC or MOSEI, and a robust, directionally consistent degradation on FI personality despite FI carrying strong real graph signal by every measure we computed — the router's failure on FI is not signal absence, and the picture is more specific than a uniform "signal absent or unused" story (main paper, Section 4; §).

Leakage Protocol Details

Graph Construction per Split

For each dataset, we construct KNN graphs using one of three leakage-safe modes (see Section 3, "Method," of the main paper):

- Inductive: the graph is built on training-split embeddings only. Validation and test nodes connect exclusively to their K nearest training neighbours; they never connect to each other.
- Split-local: separate graphs are built independently for train, validation, and test. No edge crosses a split boundary.
- Transductive (ablation only, not leakage-safe): a single graph is built over all splits jointly, so validation/test nodes may connect to each other. Retained only to quantify how much a naive (non-leakage-safe) protocol would inflate results.

DAIC-WOZ: 189 sessions (107 train / 35 val / 47 test), subject-independent — no participant ID appears in more than one split. CMU-MOSEI and ChaLearn FI use their official utterance-/clip-level splits (see Section 3, "Method," of the main paper). All splits are validated by an automated leakage audit (9/9 checks passing: inductive train/test, split-local val/test, transductive cross-split-edges-expected, bug-injection detection, and LLM-feature independence).

Table 1: Complete hyperparameter configuration.

Parameter	Value
Common dimension d	256
MMoEEx experts M	8
Expert hidden size (both configurations)	256
GraphSAGE layers	2
GraphSAGE hidden size	126
GraphSAGE aggregation	Mean
GAT heads / head dim (ablation)	3 / 42
KNN neighbours K	10 or 15 (per variant)
Edge similarity metric	Cosine
Graph routing weight λ	0.5 (fixed, not tuned)
Learning rate	3×10^{-4}
Optimizer	AdamW
Batch size	32
Epochs (max)	150
Early stopping patience	20
Projector freeze epochs	20
Temperature balance T	3.0

Hyperparameters

Complete Ablation Results

Row 1's DAIC AUROC (0.584) is a same-protocol recomputation (nested-CV ℓ_2-regularized logistic regression on the raw 768-dim RoBERTa embedding, identical train/test participants as every other DAIC number in this paper), not the different value an earlier project phase reported for the same nominal quantity. That earlier number could not be reproduced from its own unmodified code on the features currently on disk; the discrepancy is disclosed in full in § and is not used as a baseline or sanity check anywhere in this paper.

Rows 3–6 are averaged over the 5 canonical seeds; earlier versions of this table (and of the main paper) reported a single-seed non-graph baseline of 0.573, a single-seed KNN-voting DAIC AUROC of 0.630, and

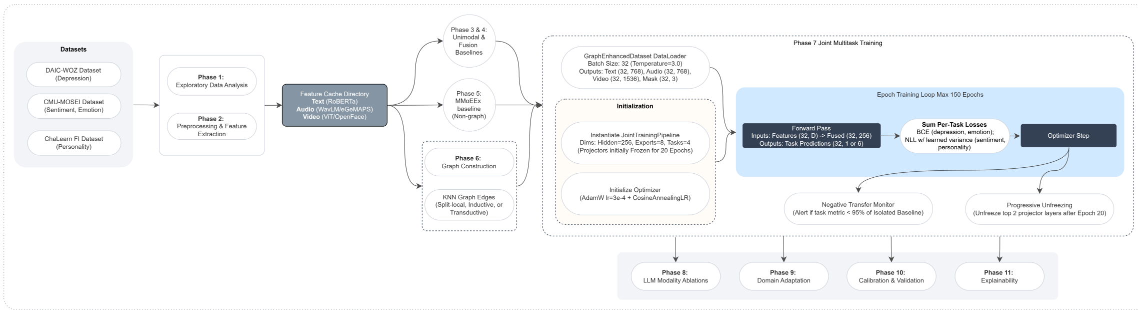

Figure 1: Implementation detail of the experimental pipeline: the non-graph MMoEEx baseline (Phase 5) and the graph-routed joint multitask training (Phase 7) share the same DataLoader, initialization, and epoch training loop, differing only in whether the graph router contributes to expert routing; both feed the central-result analysis (construct-profile comparison, MPDD dissociation, 5-seed reportability statistics) and the claim-verification harness. LLM ablations, domain adaptation, calibration, and explainability (dashed) are single-seed exploratory material, excluded from the paper's claims.

Table 2: Full ablation ladder. Rows 0–1 are single-seed (not yet extended to the 5-seed protocol used elsewhere in this paper; flagged accordingly). Rows 3–6 report mean ± std over the 5 canonical seeds (17, 42, 1337, 2024, 31415). An intermediate row present in earlier drafts (gated late fusion, no MMoEEx) is not reported here; it could not be reproduced and showed diagnostic signs of model collapse (§). Row numbering is preserved from that earlier draft for cross-reference. Source: artifacts/stats/routing_table1_5seed.json, artifacts/stats/knn_voting_5seed.json, artifacts/tables/unimodal_baselines.csv.

		DAIC AUROC	MOSEI Sent. CCC	MOSEI Emotion AUROC	FI Avg CCC
0	Trivial (majority class)‡	0.500	0.000	0.500	0.000
1	+ Unimodal (best modality: text/text/video)‡	0.584	0.512	–†	0.458
3	+ MMoEEx (no graph)	0.541 ± 0.017	0.482 ± 0.011	0.710 ± 0.005	0.571 ± 0.006
4	+ KNN voting (no learned router)	0.623 ± 0.071	0.428 ± 0.021	0.674 ± 0.013	0.373 ± 0.011
5	+ Graph router (V0, inductive K=10)	0.518 ± 0.027	0.510 ± 0.024	0.722 ± 0.009	0.480 ± 0.060
6	+ Graph router (V3, inductive K=15)	0.549 ± 0.057	0.501 ± 0.022	0.714 ± 0.014	0.507 ± 0.047

† Per-modality unimodal baselines evaluate MOSEI emotion as a regression task (CCC per emotion, averaged); no per-modality classification AUROC was computed at the unimodal stage, so this cell is not directly comparable to the classification AUROC reported for rows 3–6 and is left blank rather than filled with an unrelated number. ‡ Single-seed; not yet extended to the 5-seed protocol.

single-seed graph-router values of 0.489 (V0) and 0.533 (V3), which together read as "KNN voting nearly matches the best-in-ladder baseline, and graph routing underperforms baseline on every configuration." Under the 5-seed reportability-gated comparison (main paper, Section 4; SPEC-H4-03), none of the DAIC deltas between rows 3–6 are distinguishable from seed noise at n=5 — including row 4's apparent DAIC advantage over row 3 (+0.083, paired p=0.087, not reportable); the single-seed table's directional story does not survive proper multi-seed treatment and is retracted for DAIC. Row 4 does, however, show a reportable, robust negative delta from the non-graph baseline on all three other metrics (MOSEI sentiment CCC p=0.003, MOSEI emotion AUROC p=0.014, FI CCC p<0.001; artifacts/stats/knn_voting_vs_baseline_statistics.json) — simple neighborhood label-smoothing is not a free lunch once measured across seeds, even though it happens to land numerically close to the DAIC baseline. FI personality is the one metric where the learned graph-router rows (5–6) also show a reportable, robust negative delta from the non-graph baseline (main paper Table 1; 3 of 5 variants significant at $p < 0.05$, the other two borderline) — consistent with, not contradicted by, row 4's own FI degradation.

Fusion ablation (single-seed). An early, single-seed fusion comparison (Phase 4, a separate isolated-fusion training run rather than the joint multitask pipeline used everywhere else in this paper) found cross-attention fusion below gated late fusion on DAIC AUROC (0.379 vs. 0.442) and MOSEI sentiment CCC (0.540 vs. 0.575); on FI, both fusion types collapsed to near-zero CCC in this isolated ablation and the comparison is uninformative there. This was never extended to the 5-seed protocol and is reported here, single-seed and exploratory, rather than in the main paper's Results.

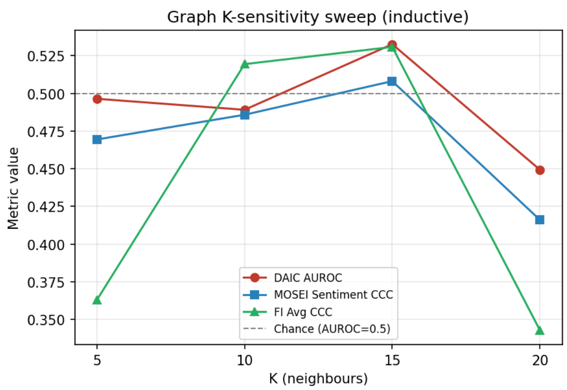

Figure 2: 5-seed mean $\pm$ std, inductive graph: no monotonic trend with $K \in \{5, 10, 15, 20\}$ (K=10, K=15 are V0/V3 above). DAIC AUROC: 0.541 ± 0.048 (K=5), 0.518 ± 0.027 (K=10), 0.549 ± 0.057 (K=15), 0.526 ± 0.057 (K=20) — none reportably different from the non-graph baseline (0.541 ± 0.017) under SPEC-H4-03. An earlier single-seed version of this sweep (one training run per K: 0.496, 0.489, 0.533, 0.449) read as showing no monotonic trend either, but with materially different point estimates at every K; the qualitative conclusion (no configuration recovers a DAIC benefit) is unchanged under 5-seed treatment, though the specific single-seed values do not reproduce.

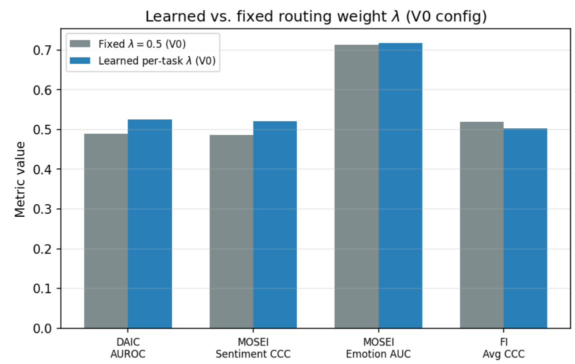

Figure 3: 5-seed mean $\pm$ std: learned per-task λ (V0 configuration) reaches DAIC AUROC 0.546 ± 0.034 against the fixed-λ=0.5 V0 baseline's 0.518 ± 0.027 — a delta of +0.028 that is not reportable under SPEC-H4-03's stricter effect-size criterion, despite a naive paired t-test alone giving p=0.007. FI CCC moves from 0.480 ± 0.060 (fixed λ) to 0.501 ± 0.047 (learned), also not reportable. A single-seed comparison (one training run each) previously read learned λ as producing "a real gain on DAIC" (AUROC 0.489 → 0.525); Table 4 documents this as one of six such reversals.

Graph-Sensitivity Sweep

K=5 shows a reportable, robust negative FI CCC delta from the non-graph baseline (0.467 ± 0.074 vs. 0.571 ± 0.006, paired p=0.04), consistent with the FI-degradation pattern in Table 1; K=20's FI delta is borderline (p=0.06), matching V3/V4's borderline status at K=15.

Learned Routing Weight

Additional Figures

Single-Seed Exploratory Material: LLM Encoders, Calibration, Domain Adaptation, Explainability

No claims are made from this section. Every result below was run on a single seed, was never extended to this paper's 5-seed protocol, and touches none of the paper's three evidenced claims (the construct-projection inversion, the MPDD dissociation, or the localizability/homophily result). We include it because it was part of this project's development history, not because it supports or contradicts anything in the main paper. Table 3 summarizes it; no number here should be cited as a finding.

Expert Routing Analysis

Figure 5 shows the real MMoEEx routing-weight distribution (non-graph baseline): each task concentrates weight on a small subset of its assigned experts rather than routing uniformly, indicating the gates learn task-differentiated behaviour even where the aggregate metric for that task declines relative to a unimodal baseline (see main paper).

Explainability Case Studies

Beyond the routing-performance question, the main paper's Introduction motivates two further, separately-evaluated hypotheses: that mental health assessment can be made more transparent by characterizing depression jointly with sentiment, emotion, and personality rather than in isolation, and that a graph over similar individuals can give both a visual and a machine-readable account of what informed a prediction. This section reports both.

Multi-Construct Characterization

For 9 case studies (3 per dataset), we pass the same sample's shared representation through all four task heads — not only the head matching its own source dataset — producing a depression probability, sentiment score, six emotion probabilities, and five personality-trait scores for the same person in one forward pass. On a representative positive-sentiment MOSEI case, the profile is internally coherent: happiness probability 0.89, sentiment score +0.62, with sadness/anger/fear all below 0.09 — a plausible joint characterization rather than four unrelated numbers. Related work explicitly separating personality-trait context from acute depression symptoms in an interpretable graph model (Vyalla et al. 2026) supports this direction; each pro-

Table 3: Single-seed exploratory results, not extended to this paper's 5-seed protocol, reported for development-history completeness only. None of these touch the main paper's evidenced claims.

Analysis	Single-seed result (exploratory only)
LLM-based encoders (L0–L5)	Replacing classical encoders with Mistral-7B/CLAP/LLaVA-1.5-7B: point-estimate gains on MOSEI (sentiment CCC +0.140, emotion AUROC +0.064), a gain on DAIC only from the full LLM stack (+0.076), and a loss on FI at every level. In the original single-seed analysis, no DAIC comparison against the classical baseline reached significance by the DeLong test (all $p > 0.72$, n=35) — these were already reported as suggestive, not confirmed, before this paper's 5-seed protocol existed.
Post-hoc calibration	Temperature/Platt/isotonic scaling on the DAIC classifier (single LLM level): isotonic regression reduced ECE most (0.26–0.35 uncalibrated to lower after calibration); all three methods are monotonic and leave AUROC-based discrimination unchanged by construction.
Cross-dataset domain adaptation	CORAL/MMD/DANN/combined, transferring FI/MOSEI representations to DAIC: underperformed a no-adaptation baseline in 10 of 12 conditions, single seed. Separately, on the external MPDD benchmark (Wav2Vec2+OpenFace features, distinct from this paper's MPDD ground-truth-vs-estimated dissociation, main paper Section 4): a regularized logistic regression outperformed our architecture (AUROC 0.674 vs. 0.545–0.559); zero-shot Young→Elderly transfer collapsed below chance (0.395); MPDD→DAIC transfer was positive (0.551).
Explainability case studies	9 case studies (3 per dataset) passing each sample through all four task heads, paired with SHAP/GNNExplainer/GraphXAIN-style post-hoc narratives. A faithfulness check (modalities outside a dataset's native routing show exactly zero attribution, 43/90 perturbation tests) held up; rebuilding the explanation graph on the model's own trained representation did not improve DAIC label homophily (0.463 trained vs. 0.537 random vs. 0.577 raw, on 35 samples) and LLM-generated narratives were observed to invent clinical detail not present in their prompt.

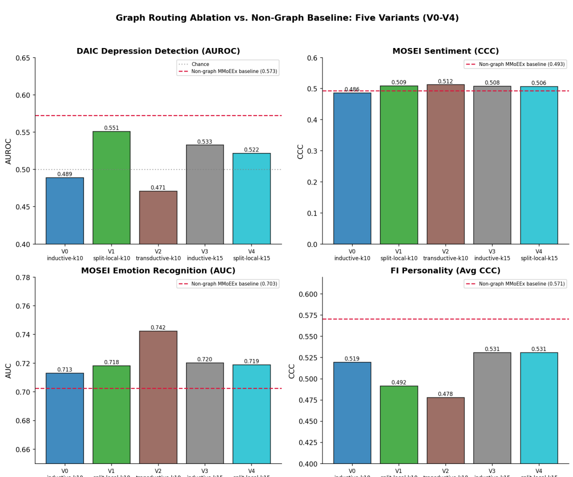

Figure 4: Graph routing ablation vs. non-graph MMoEEx baseline (red dashed line), all five variants across all four tasks.

file is stored as a multi_construct_profile field in artifacts/tables/phase11_xai_results.json.

Graph-Based Explanation

For each case study we generate a visual record (SHAP per-modality beeswarm plot, GNNExplainer-style subgraph diagram (Ying et al. 2019), and a GraphXAIN-style (Cedro and Martens 2024) natural-language nar-

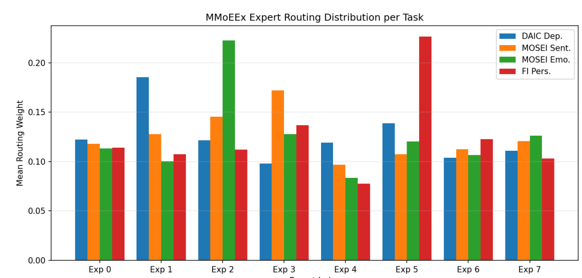

Figure 5: Mean MMoEEx routing weight per expert and task (non-graph baseline). Rows/legend are tasks; columns are the 8 experts.

rative panel, Figure 6) and a machine-readable record (SHAP values, top-k neighbour identities and cosine distances, expert routing weights, the multi-construct profile, and counterfactual/perturbation deltas). The narrative is generated by prompting an LLM (Mistral-7B) with the SHAP values, subgraph structure, and sample metadata, used purely as post-hoc explanation of an independently-computed prediction — not as reasoning inserted into the prediction pipeline itself. This distinction matters: chain-of-thought-style reasoning improves mental-health text classification when used to produce the prediction (Patil and Gedhu 2025), but can be unfaithful or reduce reliability when embedded in the clinical decision itself (Wu et al. 2025).

Every SHAP, perturbation, and counterfactual value

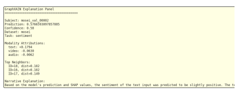

Figure 6: GraphXAIN-style narrative panel for a representative MOSEI case, combining SHAP attribution, subgraph structure, and sample metadata into a post-hoc natural-language explanation.

must be computed under each sample's own dataset-native routing (main paper, Section 3): DAIC is evaluated text-only and FI video-only, so a modality outside a dataset's native routing shows exactly zero attribution, not a small residual. Across 90 perturbation tests, only 47 are non-zero, exactly matching each sample's routing; the remaining 43 are exactly zero rather than merely small. This is a stronger faithfulness check than a ranking comparison: it directly confirms the explanation pipeline reflects each dataset's real computation.

The explanation graph is built on the model's own trained fused representation rather than raw concatenated features, so reported neighbours are the ones the router and task heads actually operate on. This does not, however, improve DAIC label homophily: on the same 35 DAIC validation samples used for the main paper's homophily diagnostic, label agreement is 0.577 for raw features, 0.537 for random pairing, and 0.463 (below random) for the trained representation, since task-supervised training optimizes for the final decision boundary rather than for preserving neighbourhood structure. DAIC subgraph explanations should be read as faithful to the model's own computation, without a validated claim that the shown neighbours are similar in a depression-relevant sense; MOSEI and FI do not share this limitation, consistent with the main paper's homophily diagnostic finding real label signal for those tasks.

Reproducibility Appendix: Documented Discrepancies and Failure Modes

This section discloses numeric discrepancies surfaced during this work's own verification harness, reported here rather than silently resolved or omitted, per the harness's own quarantine protocol: a number under investigation may be disclosed only in this section until its status is settled.

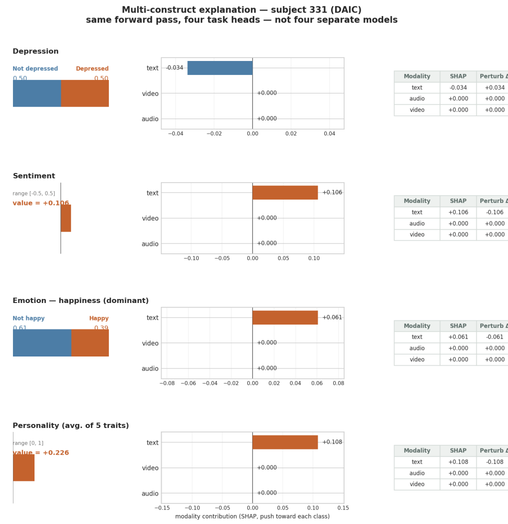

Figure 7: Multi-construct force plot: one DAIC subject explained through all four task heads in a single figure. Audio and video read exactly zero on every row, visualizing DAIC's text-only routing directly rather than leaving it implicit.

Six Cases Where Methodology Changed the Answer

The central claim of this paper is that single-seed, unverified results in this class of pipeline can mislead. Table 4 documents six concrete instances of this, all from this project's own history, each caught by an automated check rather than manual review.

We report this catalogue as an argument for the harness rather than an embarrassment: every one of these six would have been a false or overstated claim in the manuscript had it not been checked, and every check that caught one was cheap relative to the cost of an incorrect published claim.

Unresolved: Phase 3 Text-Only AUROC Discrepancy

An earlier project phase (Phase 3, prior to this paper's harness) reported DAIC text-only AUROC as 0.671, computed by a bespoke script with its own logistic-regression configuration. This paper's claim-verification harness recomputes the identical nominal quantity — text-only DAIC depression decoding, same participants, same train/test split — under one consistent nested-cross-validated protocol used for every number in this paper, obtaining 0.584. Attempting to close the gap, we ruled out two candidate explanations: (i) feature loading — both pipelines read the identical cached RoBERTa embeddings from the identical manifest, confirmed by directly invoking the earlier script's own unmodified data-loading code in this session; (ii) classifier hyperparameters — we swept all four combinations of (fixed C=1 vs. cross-validated C) × (class-balanced

Table 4: Six cases from this project where a number changed, or was retracted, once checked. Each was caught by an automated mechanism (a generalization gate, a leakage/consistency audit, a multi-seed reportability rule, or a same-protocol recomputation), not by manual re-reading.

Case	Before	After
MPDD personality-decoding inversion	AUROC 0.174 (predicted profile), computed before checking the regressor's held-out R^2	Discarded: regressor $R^2 < 0$ on held-out data (train R^2 0.91–0.99); the number reflected overfit-model noise, not a real effect (§ above)
In-domain DAIC control	Single checkpoint: delta -0.108, read as corroborating the cross-corpus inversion	5 seeds: delta -0.037 ± 0.084, 3 of 5 negative, 2 of 5 positive, sign test p=0.5 — inconclusive, not corroborating (main paper, Section 4)
KNN-voting DAIC advantage	Single seed: AUROC 0.630, read as "best DAIC AUROC in the ladder except the non-graph baseline"	5 seeds: +0.083 over baseline, paired p=0.087 — not reportable; reportably worse than baseline on all three other metrics
Non-graph MMoEEx baseline	DAIC AUROC 0.493 for a full submission cycle, read as a working classifier	Chance-level: a per-sample sampling weight was computed but never passed to the training DataLoader; fixed, reaches 0.573 (single seed) / 0.541 ± 0.017 (5 seeds)
Learned per-task λ	Single seed: DAIC AUROC 0.489 → 0.525, read as "a real gain on DAIC" from replacing the fixed λ=0.5	5 seeds: delta +0.028 vs. fixed-λ V0, not reportable under SPEC-H4-03's stricter effect-size criterion despite naive paired p=0.007
Phase 3 text-only AUROC	0.671, used as a sanity anchor for the corrected baseline	Irreproducible: re-running the original script's own unmodified code on current features gives 0.515, not 0.671; same-protocol recomputation gives 0.584 (§ above)

vs. unbalanced) on the identical feature arrays and reproduced neither 0.671 nor anything close to it (range: 0.515–0.584). Re-running the earlier phase's own unmodified code, on the features currently on disk, reproduces 0.515, not its own previously published 0.671 — a concrete instance, within this project, of exactly the failure mode this paper's construct-estimation results describe: a number that cannot be regenerated from its own stated method is not a fact about the world, whatever it once measured. Root cause remains undetermined (most likely an unlogged upstream feature-cache regeneration). 0.671 is not used as a baseline, sanity check, or comparison point anywhere else in this paper; every reported DAIC text-only figure is the recomputed 0.584.

Documented Failure Mode: An Overfit Intermediate Predictor

The main paper's construct-bottleneck analysis (Section 4) includes an MPDD arm that trains a ridge regressor to estimate Big-Five personality scores from that corpus's own raw audio/video features, as a construct-supervised counterpart to the DAIC profile. Before using this regressor's predictions in any downstream comparison, we checked its held-out generalization: training-set R^2 was 0.91–0.99 across the five traits, but held-out test-set R^2 was negative for four of five (worst: -2.86; only Agreeableness reached a modest positive value, $R^2 \approx 0.1$, and only at some PCA-reduced dimensionalities we tried). This is textbook overfitting at this feature-to-sample ratio (1221 raw features, 184 training subjects).

Before this check was run, an initial version of the downstream comparison — depression decoding from this regressor's predicted trait profile, against a matched-dimensionality random projection of the same raw features — produced a striking result: predicted-profile AUROC of 0.174 (i.e., a strong inversion, well below chance), against a random-projection AUROC of 0.590. Reported without the generalization check, this would have read as a dramatic confirmatory instance of this paper's central claim. It is not one. A predictor with negative held-out R^2 is not a valid construct estimate at all; its downstream AUROC reflects properties of an overfit model's residual error structure, not a real relationship between constructs and depression. We discarded this number rather than reporting it, once the R^2 check surfaced the problem, and it is not used as evidence for any claim in this paper.

We record this as a general methodological finding as much as a project-specific one: an overfit intermediate predictor, at feature-to-sample ratios that are common in small-cohort multimodal mental-health work, can manufacture a striking, publishable-looking downstream effect out of noise, in either direction. We adopt as a standing rule for this project — and recommend as general practice — that any derived predictor whose output feeds a downstream analysis must clear a held-out generalization check before its output may be consumed; the in-domain DAIC sentiment/emotion re-derivation in the main paper's construct-bottleneck analysis (Section 4) applies the same rule and reports, for that case, which dimensions passed and which did not.

Diagnostic Note: An Irreproducible, Degenerate Ablation Row

An earlier version of the ablation ladder (Table 2) included a "gated late fusion, no MMoEEx" row reporting DAIC AUROC 0.442, MOSEI sentiment CCC 0.575, MOSEI emotion AUROC 0.634, and FI CCC exactly 0.000. We could not identify a current script whose configuration matches this row's description well enough to rerun it on request, and an FI CCC of exactly 0.000 is itself a constant-output collapse that this project's own generalization checks would flag rather than report as a working result. Per this appendix's standing policy that a number a) cannot be regenerated from stated method and b) shows diagnostic signs of model collapse is not a fact about the world, we have removed this row from the ladder rather than report it with a caveat; 0.442 does not appear in any table in this paper.

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